7.1.3 The Second Variation

[43] Before we apply the calculus of variations to problems in classical mechanics, there is one thread that we left hanging. We know that the Euler–Lagrange equation is the requirement for a function to be an extremum. But how do we tell if it's a maximum or minimum or something else? It turns out that this information won't actually be needed for our applications to classical mechanics, but we include it here for completeness.

[44] To set the scene, we again return to a function of several variables $f(y_1, \ldots, y_n)$. Taylor expanding about a given point **y**, we have

$$f(\mathbf{y} + \delta\mathbf{y}) - f(\mathbf{y}) = \frac{\partial f}{\partial y_i}(\mathbf{y})\,\delta y_i + \frac{1}{2}\frac{\partial^2 f}{\partial y_i \partial y_j}(\mathbf{y})\,\delta y_i \delta y_j + \ldots . \tag{7.28}$$

複習註解：7.28 式的兩個簡單例子

7.28 式是多變數函數的二階 Taylor 展開：

$$f(\mathbf{y} + \delta\mathbf{y}) - f(\mathbf{y}) = \frac{\partial f}{\partial y_i}\delta y_i + \frac{1}{2}\frac{\partial^2 f}{\partial y_i \partial y_j}\delta y_i \delta y_j + \cdots .$$

例 **1**：$f(x, y) = x^2 + y^2$
取展開點 $(1, 2)$。有

$$\nabla f = (2x, 2y),$$

所以在 $(1, 2)$，

$$\nabla f = (2, 4).$$

Hessian matrix 為

$$H = \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}.$$

因此

$$\Delta f = 2\delta x + 4\delta y + \delta x^2 + \delta y^2.$$

這裡一階項描述主要變化方向，二階項描述曲率。

例 **2**：$f(x, y) = x^2 - y^2$
取 stationary point $(0, 0)$。因為

$$\nabla f = (2x, -2y),$$

所以在 $(0, 0)$，

$$\nabla f = (0, 0).$$

因此一階變化消失，只剩二階項。Hessian matrix 為

$$H = \begin{pmatrix} 2 & 0 \\ 0 & -2 \end{pmatrix},$$

所以

$$\Delta f = \delta x^2 - \delta y^2.$$

若沿 x 方向，令 $\delta y = 0$，則

$$\Delta f = \delta x^2 > 0.$$

若沿 y 方向，令 $\delta x = 0$，則

$$\Delta f = -\delta y^2 < 0.$$

因此在同一個 stationary point 附近，有些方向使函數增加，有些方向使函數減少，所以 $(0,0)$ 是鞍點，而不是極大值或極小值。

7.28 式的二階項決定 stationary point 附近的曲率，並自然導向 Hessian matrix。

複習註解：為什麼二階項可以寫成 $\frac{1}{2}\delta\mathbf{y}^T H\delta\mathbf{y}$？

這不是形式上剛好可以這樣寫，而是由 7.28 式中的二階項按照矩陣乘法系統整理而來。
7.28 式的二階項為

$$\frac{1}{2}\frac{\partial^2 f}{\partial y_i \partial y_j}\delta y_i \delta y_j.$$

其中重複指標 i, j 表示要對所有 i, j 加總，因此更完整地寫成

$$\frac{1}{2}\sum_{i=1}^{n}\sum_{j=1}^{n}\frac{\partial^2 f}{\partial y_i \partial y_j}\delta y_i \delta y_j.$$

定義 Hessian matrix

$$H_{ij} = \frac{\partial^2 f}{\partial y_i \partial y_j},$$

以及擾動向量

$$\delta\mathbf{y} = \begin{pmatrix} \delta y_1 \\ \delta y_2 \\ \vdots \\ \delta y_n \end{pmatrix}.$$

依照矩陣乘法，

$$\delta\mathbf{y}^T H\delta\mathbf{y} = \sum_{i=1}^{n}\sum_{j=1}^{n}\delta y_i H_{ij}\delta y_j.$$

代入 H_{ij} 的定義便得到

$$\delta\mathbf{y}^T H\delta\mathbf{y} = \sum_{i,j}\frac{\partial^2 f}{\partial y_i \partial y_j}\delta y_i \delta y_j.$$

因此二階 Taylor 項可以寫成

$$\boxed{\Delta f \approx \frac{1}{2}\delta\mathbf{y}^T H\delta\mathbf{y}.}$$

以二變數為例，

$$H = \begin{pmatrix} f_{xx} & f_{xy} \\ f_{yx} & f_{yy} \end{pmatrix}, \qquad \delta\mathbf{y} = \begin{pmatrix} \delta x \\ \delta y \end{pmatrix},$$

則

$$\delta\mathbf{y}^T H \delta\mathbf{y} = f_{xx}(\delta x)^2 + f_{xy}\delta x\delta y + f_{yx}\delta y\delta x + f_{yy}(\delta y)^2.$$

若 $f \in C^2$，則 $f_{xy} = f_{yx}$，所以

$$\delta\mathbf{y}^T H \delta\mathbf{y} = f_{xx}(\delta x)^2 + 2f_{xy}\delta x\delta y + f_{yy}(\delta y)^2,$$

正好就是多變數 Taylor 展開的二階項。
因此可以記成

$$\boxed{\text{二階偏導的雙重求和} \iff \text{Hessian 所形成的矩陣二次型.}}$$

把二階項寫成矩陣形式的好處，是後面可以直接利用對稱矩陣的譜定理，將 Hessian 正交對角化，再由特徵值的正負判斷 stationary point 的性質。

[45] We assume that $\mathbf{y}$ is a stationary point, so $\partial f/\partial y_i(\mathbf{y}) = 0$. That leaves us with the second order variation which is determined by the *Hessian* matrix

$$H_{ij}(\mathbf{y}) = \frac{\partial^2 f}{\partial y_i \partial y_j}(\mathbf{y}). \tag{7.29}$$

複習註解一：stationary point（駐點）

駐點表示「一階變化為零」。對一般函數，條件為 $f'(x_0) = 0$；對多變數函數，則為 $\nabla f = 0$。在變分法中，駐點是一條使泛函的一階變分對所有容許擾動皆為零的路徑：

$$\delta F[y] = 0.$$

這不代表路徑的斜率 $y' = 0$，也不保證是極大值或極小值；還需要用第二變分判斷。

複習註解二：Hessian 矩陣如何描述二階變化？

Hessian 矩陣把所有二階偏導數排列在一起，扮演一變數函數中 f'' 的角色。以兩變數為例，

$$H = \begin{pmatrix} \dfrac{\partial^2 f}{\partial y_1^2} & \dfrac{\partial^2 f}{\partial y_1 \partial y_2} \\ \dfrac{\partial^2 f}{\partial y_2 \partial y_1} & \dfrac{\partial^2 f}{\partial y_2^2} \end{pmatrix}.$$

對角元素描述各座標方向的二階彎曲；混合偏導數描述兩個變數一起改變時的交互影響。若二階偏導數連續，混合偏導數相等，故 $H^T = H$。在駐點 $\mathbf{y}_0$，一階項消失，泰勒展開為

$$f(\mathbf{y}_0 + \delta\mathbf{y}) - f(\mathbf{y}_0) = \frac{1}{2}\delta\mathbf{y}^T H(\mathbf{y}_0)\delta\mathbf{y} + o(\|\delta\mathbf{y}\|^2).$$

其中

$$\delta\mathbf{y}^T H\delta\mathbf{y} = H_{11}(\delta y_1)^2 + 2H_{12}\delta y_1\delta y_2 + H_{22}(\delta y_2)^2.$$

因此，判斷駐點要看所有方向的二階變化，不能只看對角元素。特徵值全正時為嚴格局部極小值；全負時為嚴格局部極大值；正負並存時為鞍點。若含零特徵值且沒有正負並存，二階資訊不足，需進一步分析。

[46] Because this matrix is symmetric, it can always be diagonalised. This means that there exists a rotation matrix $R \in O(n)$ such that $H' = R^T H R$ is diagonal, with the elements given by the eigenvalues λ_i of H

$$H' = \text{diag}(\lambda_1, \ldots, \lambda_n). \tag{7.30}$$

[47] The nature of the stationary point is determined by the signs of these eigenvalues:

- If $\lambda_i > 0$ for all $i = 1, \ldots, n$, then the stationary point is a minimum.
- If $\lambda_i < 0$ for all $i = 1, \ldots, n$, then the stationary point is a maximum.
- If λ_i are a collection of positive and negative numbers then the stationary point is a saddle point.
- If some $\lambda_i = 0$ then we have to work harder and look to higher orders to understand what's going on in those directions.

複習註解三：Hessian 的兩個例子

例一：$f(y_1, y_2) = y_1^2 + 3y_2^2$。由 $\nabla f = (2y_1, 6y_2)^T = 0$，駐點為 $(0, 0)$，且

$$H = \begin{pmatrix} 2 & 0 \\ 0 & 6 \end{pmatrix}, \qquad \Delta f = \frac{1}{2}\delta\mathbf{y}^T H\delta\mathbf{y} = (\delta y_1)^2 + 3(\delta y_2)^2.$$

任何非零位移都讓函數值增加，所以原點是嚴格局部極小值，本例也是全域極小值。特徵值 $2, 6$ 皆為正；沿 y_2 方向的二階彎曲較強。

例二：$f(y_1, y_2) = y_1^2 + 4y_1y_2 + y_2^2$。由 $2y_1 + 4y_2 = 0$ 與 $4y_1 + 2y_2 = 0$，駐點也是 $(0, 0)$，且

$$H = \begin{pmatrix} 2 & 4 \\ 4 & 2 \end{pmatrix}, \qquad \Delta f = (\delta y_1)^2 + 4\delta y_1\delta y_2 + (\delta y_2)^2.$$

雖然兩個對角元素皆為正，但沿不同方向移動會得到不同符號：

$$\begin{aligned} \delta\mathbf{y} = (t, t)^T \quad &\Longrightarrow \quad \Delta f = 6t^2 > 0, \\ \delta\mathbf{y} = (t, -t)^T \quad &\Longrightarrow \quad \Delta f = -2t^2 < 0 \end{aligned} \qquad (t \neq 0).$$

因此原點是鞍點。由

$$\det(H - \lambda I) = (2 - \lambda)^2 - 16 = 0$$

可得特徵值 $6, -2$，對應特徵方向分別為 $(1, 1)^T$ 與 $(1, -1)^T$。

兩例皆為二次式，因此 $\Delta f = \frac{1}{2}\delta\mathbf{y}^T H\delta\mathbf{y}$ 是精確等式，沒有更高階項。

[48] Now we extend this to functionals $F[y(x)]$. We vary the path from $y(x)$ to $y(x) + \delta y(x)$ and write

$$\delta y(x) = \epsilon \eta(x) \tag{7.31}$$

with $\epsilon \ll 1$. We then expand the functional (7.7) to second order

$$\begin{aligned} \delta F[y(x)] = \epsilon \left\{ \int_a^b dx \left[\frac{\partial L}{\partial y} - \frac{d}{dx}\left(\frac{\partial L}{\partial y'} \right) \right] \eta(x) + \left[\frac{\partial L}{\partial y'} \eta \right]_a^b \right\} \\ + \frac{1}{2}\epsilon^2 \int_a^b dx \left\{ \frac{\partial^2 L}{\partial y^2}\eta^2 + 2\frac{\partial^2 L}{\partial y \partial y'}\eta\eta' + \frac{\partial^2 L}{\partial y'^2}\eta'^2 \right\}. \end{aligned} \tag{7.32}$$

[49] The first line is what we had before in (7.10). The second line is new. We consider paths between $x = a$ and $x = b$ with fixed end points, so that $\eta(a) = \eta(b) = 0$. Moreover, we'll take a path that solves the Euler–Lagrange equation. This means that the first line vanishes and we're left just with the second-order term.

[50] We have three terms, proportional to η^2, $\eta\eta'$, and η'^2. Unfortunately, we don't get to integrate by parts to make all of these look the same. We can, however, use the fact that $2\eta\eta' = (\eta^2)'$ to integrate the second by parts. For an extremum of the action, the second variation is then

$$\delta^2 F[y(x)] = \frac{1}{2}\epsilon^2 \int_a^b dx \left\{ \left[\frac{\partial^2 L}{\partial y^2} - \frac{d}{dx}\left(\frac{\partial^2 L}{\partial y \partial y'} \right) \right] \eta^2 + \frac{\partial^2 L}{\partial y'^2}\eta'^2 \right\}. \tag{7.33}$$

複習註解：7.33 式與二階變分的角色

對泛函

$$F[y] = \int_a^b L(y, y', x)\, dx,$$

令路徑做變化

$$y(x) \to y(x) + \epsilon\eta(x),$$

且固定端點，因此

$$\eta(a) = \eta(b) = 0.$$

前面的一階變分

$$\delta F = 0$$

給出 Euler–Lagrange 方程，它只能告訴我們 $y(x)$ 是 stationary path，但還不能判斷它是極小、極大或其他情形。

因此需要研究二階變分。

二階項原本包含

$$\eta^2, \qquad \eta\eta', \qquad (\eta')^2.$$

利用

$$2\eta\eta' = (\eta^2)'$$

對混合項做分部積分，並利用固定端點條件 $\eta(a)=\eta(b)=0$ 消去邊界項，可整理成

$$\delta^2 F[y]=\frac{1}{2}\epsilon^2\int_a^b\left\{\left[\frac{\partial^2 L}{\partial y^2}-\frac{d}{dx}\left(\frac{\partial^2 L}{\partial y\,\partial y'}\right)\right]\eta^2+\frac{\partial^2 L}{\partial y'^2}(\eta')^2\right\}dx.$$

這就是泛函版本的二次型。
普通多變數函數中，我們研究

$$\delta\mathbf{y}^T H\delta\mathbf{y};$$

在變分法中，對應的角色則由

$$\delta^2 F[y]$$

扮演。
因此可以記成

Euler–Lagrange 方程找 stationary path，二階變分判斷 stationary path 的性質。

[51] Here we've thrown away the boundary term on the grounds that $\eta(a)=\eta(b)=0$. We've also renamed this term $\delta^2 F$ to emphasise that it is the second-order variation.

[52] The variation (7.33) depends both on the path $y(x)$ that solves the Euler–Lagrange equation, and also on the variation $\eta(x)$. At this point, the story is a more complicated version of what we saw for a function of several variables. Suppose, for example, that, when evaluated on our extremum path $y(x)$, the second-order variation (7.33) is strictly positive:

$$\delta^2 F[y(x)]>0\quad\text{for all }\eta(x). \tag{7.34}$$

[53] In this case, our extremum was a minimum of the functional. However, it may not be so easy to check that (7.34) holds because we have to do it for all functions $\eta(x)$. There is, however, a simple test that we can perform. If the path $y(x)$ is a local minimum, then it necessarily obeys

$$\frac{\partial^2 L}{\partial y'^2}\geq 0\quad\text{for all }x. \tag{7.35}$$

[54] This is known as the *Legendre condition*. We recognise $\partial^2 L/\partial y'^2$ as the final term in (7.33). The idea behind the Legendre condition is that if (7.35) does *not* hold, so that $\partial^2 L/\partial y'^2<0$ for some region, then we can always find a test function so that η is small in that region while η' is large and make the final term in (7.33) arbitrarily negative so that $\delta^2 F<0$. The Legendre condition is therefore necessary for the path to be a minimum. It is not, however, sufficient.

複習註解：為什麼 **Legendre condition** 只是必要條件？

若 $y(x)$ 真的是泛函的局部極小路徑，則對所有非零變分函數 $\eta(x)$，應滿足

$$\delta^2 F[y]>0.$$

但直接檢查所有可能的 $\eta(x)$ 通常很困難，因此作者提出較簡單的必要條件

$$\boxed{\frac{\partial^2 L}{\partial y'^2} \geq 0.}$$

原因是 7.33 式中有一項

$$\frac{\partial^2 L}{\partial y'^2}(\eta')^2.$$

由於

$$(\eta')^2 \geq 0,$$

如果在某一區域內

$$\frac{\partial^2 L}{\partial y'^2} < 0,$$

就可以選擇一個變分函數 $\eta(x)$，使該區域內 η 很小但 η' 很大。
此時含有 η^2 的項可以很小，而

$$\frac{\partial^2 L}{\partial y'^2}(\eta')^2$$

會成為主要貢獻，並使

$$\delta^2 F[y] < 0,$$

這和局部極小值所需要的二階變分非負相矛盾。
因此若 $y(x)$ 是局部極小路徑，至少必須有

$$\frac{\partial^2 L}{\partial y'^2} \geq 0.$$

但這只檢查了 7.33 式中的一部分。完整二階變分還包含

$$\left[\frac{\partial^2 L}{\partial y^2} - \frac{d}{dx}\left(\frac{\partial^2 L}{\partial y\,\partial y'}\right)\right]\eta^2.$$

所以即使 Legendre condition 成立，也不能保證

$$\delta^2 F[y] > 0$$

對所有 η 都成立。
因此

Legendre condition 是局部極小的必要條件，但不是充分條件。

複習註解：這一段真正要掌握的重點

這一段最重要的是把「普通函數的二階判別」推廣到泛函。
對泛函

$$F[y],$$

Euler–Lagrange 方程來自一階變分

$$\delta F = 0,$$

它只能告訴我們 $y(x)$ 是 stationary path，但不能直接判斷它是極小、極大或其他情形。
因此還要研究二階變分

$$\delta^2 F[y].$$

對局部極小值而言，理想情況是對所有非零變分函數 $\eta(x)$ 都有

$$\delta^2 F[y] > 0.$$

所以可以把邏輯記成

$$\boxed{\delta F = 0 \Longrightarrow \text{找到 stationary path}}$$

以及

$$\boxed{\delta^2 F \Longrightarrow \text{判斷 stationary path 的性質}}.$$

7.33 式把二階變分整理成

$$\delta^2 F[y] = \frac{1}{2}\epsilon^2 \int_a^b \left\{ A(x)\eta^2 + B(x)(\eta')^2 \right\} dx,$$

其中

$$A(x) = \frac{\partial^2 L}{\partial y^2} - \frac{d}{dx}\left(\frac{\partial^2 L}{\partial y\,\partial y'}\right), \qquad B(x) = \frac{\partial^2 L}{\partial y'^2}.$$

真正要判斷的是整個二階變分對所有 $\eta(x)$ 的正負，不是只看某一個二階偏導數。
Legendre condition

$$\boxed{\frac{\partial^2 L}{\partial y'^2} \geq 0}$$

只是局部極小值的必要條件之一。
它的意思是：若某處

$$\frac{\partial^2 L}{\partial y'^2} < 0,$$

就可以選擇一個變分函數，使 η 很小但 η' 很大，讓 $(\eta')^2$ 那一項主導並使

$$\delta^2 F[y] < 0,$$

因此不可能是局部極小值。
但即使 Legendre condition 成立，也不能保證整個

$$\delta^2 F[y]$$

一定為正，因為還要考慮 $A(x)\eta^2$ 那一項。

所以最精簡地記：

Euler–Lagrange 找駐定路徑，二階變分判斷性質，Legendre condition 是極小值的必要條件。